

KARTIK JOSHI

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SUMMARY

AI Engineer with hands-on experience across graph deep learning, reinforcement learning, and real-time systems, focused on building models that make it into working use rather than staying at the prototype stage. A Master's candidate in AI with Business, pairing applied engineering with a practical grounding in how AI supports real business decisions.

EDUCATION

SP Jain School of Global Management Dubai, UAE
Master's in AI with Business, GPA: 3.6/4.0 2025 – 2027

- Dean's List (top 10% of cohort).

Parul University Vadodara, India
B.Tech in Computer Science & Engineering (Specialization in AI), GPA: 8.04/10 2020 – 2024

- Selected as University Representative for the National Entrepreneurship Challenge at IIT Bombay.

EXPERIENCE

Founding AI Engineer November 2023 – July 2026
VaultxLedger Vadodara, India (Remote)

- Designed and demonstrated a real-time computer vision (ANPR) prototype in PyTorch and OpenCV for Vadodara Rural Police with sub-100ms inference latency, earning a Certificate of Appreciation from IPS Rohan Anand.
- Led a team of 6 engineers across pipeline architecture, hardware integration, and direct coordination with senior police officers, delivering the prototype without external QA support.

Startup Incubation Intern (PIERC Cohort-2) December 2023 – March 2024
PIERC Startup Incubator Vadodara, India

- Developed the go-to-market strategy and investor pitch for VaultxLedger, securing incubation support that led to the engagement with Vadodara Rural Police.

AI/ML Engineering Intern October 2021 – November 2021
CBbytes Vadodara, India

- Built a regression-based stock-price prediction model in Python using Support Vector Machines (SVM) to forecast prices from historical market data.

PROJECTS

CollusionGraph – Graph Deep Learning for Financial & Procurement Collusion Detection 2026
PyTorch, PyTorch Geometric, PyGOD, FastAPI, React, TypeScript, AWS, NVIDIA NIM

- Unified six public money-laundering and bid-rigging datasets into one domain-agnostic graph schema, screening for coordinated networks that single-record models cannot detect.
- Combined supervised GNNs (GATv2, GraphSAGE, R-GCN) and unsupervised detectors (DOMINANT, GAE) into an isotonic-calibrated ensemble, recovering 92% of planted bid-rigging cartels (vs. 12% for a non-learning baseline) on a 488K-node, 1.45M-edge unlabelled network where classical models could not run.
- Shipped it as a full product with PGExplainer and FATF/OECD motif-based explanations, a tool-grounded GenAI investigator copilot that passed a 24-question golden test, and a React/FastAPI console.

AlgoViz – Real-Time Algorithmic Trading Dashboard 2026
Python, NumPy, SciPy, scikit-learn, Streamlit, Plotly, Docker, Binance WebSocket API

- Engineered a live BTC/USDT trading dashboard on the Binance WebSocket API (50+ trades/sec), computing 9 market microstructure features including VWAP, TWAP, order book imbalance, and rolling volatility.
- Hand-coded an LSTM-Attention network in pure NumPy with real-time (<100ms) inference, and built a 7-component ensemble predictor with P&L backtesting, Docker deployment, and 45 pytest unit tests.

Project Wafa – Multilingual Customer Retention Intelligence Platform 2026
Python, FastAPI, DistilBERT, scikit-learn, TF-IDF, spaCy

- Owned the NLP and ML pipeline of a customer-retention platform for a simulated UAE bank, classifying issue type and churn risk in English, Arabic, Hindi, and Tagalog messages and fusing these signals with behavioural data into an explainable per-customer risk score.
- Served fine-tuned DistilBERT and TF-IDF classifiers through a FastAPI backend with a transparent rule-based decision layer, human review before any outreach is sent, and a fairness audit with nationality excluded as a feature.

- Designed a multi-agent reinforcement learning (MARL) system for adaptive traffic signal control on a 16-intersection Dubai grid, training decentralised Q-Learning and SARSA agents over 60,000 steps with a 6-factor reward balancing queues, throughput, and emissions.
- Built a simulator with 3 coordination protocols and a Chaos Mode for sandstorms, faulty sensors, and emergency-vehicle preemption, benchmarking policies via WSM and TOPSIS in a React dashboard with Q-value XAI.

PUBLICATIONS

- Co-designed the ML pipeline and evaluation framework for a cloud service recommendation system, reaching 70%+ classification accuracy across 6 user-defined parameters on a proprietary dataset of 1,000+ records.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS
ML Frameworks: PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, Hugging Face Transformers, spaCy, OpenCV, MediaPipe, NumPy, Pandas, SciPy
AI Methods: Deep Learning, Graph Neural Networks, Natural Language Processing (NLP), Computer Vision, Reinforcement Learning, LSTM & Attention, Deep SVDD, Explainable AI (XAI)
LLM & Generative AI: Google Gemini API, LLM Inference Pipelines, Prompt Engineering, Responsible AI, AI Chatbot Development
Engineering: FastAPI, React, Vite, SQLAlchemy, Docker, Git, GitHub Actions, REST APIs, Jupyter
Domains: Real-Time Systems, Edge Deployment, MLOps, Data Analytics, Smart City Systems, Web3 & Blockchain

ACHIEVEMENTS

- **Global Top 10** – ICC NIUM Hackathon, Web3 & Blockchain (International)
- **Finalist** – Smart India Hackathon 2023 (National)
- **Winner** – Let’s Hack India Tour, Parul University

LEADERSHIP & ACTIVITIES

Placement Committee Member, SP Jain School of Global Management, Dubai2025 – Present

- Coordinate between students and recruiting companies, supporting employer outreach and placements for the cohort.

AI Presenter & Product Demonstrator, India2023 – Present

- Present AI demos to stakeholders such as incubator panels, police officers, and international hackathon judges.

Marketing Lead & Event Head, CDC FunFest, Parul University, Vadodara2022 – 2023

- Led marketing and external communications with a cross-functional team for a festival of nearly 5,000 attendees.

Bal Sanchalak (Youth Leader), SMVS Swaminarayan Organization2017 – Present

- Mentor youth cohorts through structured leadership and facilitation programmes.

LANGUAGES

English (Professional) • Hindi (Conversational) • Gujarati (Native)